

it is better to include those frequencies now itself instead of purchasing additional instruments after five years. If there are no plans of expansion, the current requirements alone should be the deciding factor while specifying the instruments. Alternatively, you can go for instruments that can be augmented easily for future requirements by installing plug-in modules. Additional plug-ins can be procured when the actual requirement of a higher frequency range arises. This will reduce the investment cost.

An example is RF power meters with different power sensors or detectors. You can select a power sensor covering the required frequency range and, when the need arises for extra frequency ranges, different power sensor can be procured keeping the same basic power meter.

Sometimes one single instrument can do the job of many instruments. For example, level generator, frequency counter and power meter are combined and available as a single instrument. Cost-wise this is cheaper than purchasing three separate instruments. This is advantageous for small establishments where working manpower and space are limited. In mobile systems like mobile transmitters, such instruments are recommended as space is the limiting factor along with cost minimisation. The disadvantage, of course, is that the use of one instrument will block the use of other two instruments. A single failure can affect the entire measurements. So depending on your requirement, select a combined instrument or separate instruments.

Major specifications

Range. While ordering a test instrument, you should specify the required measuring range of the instrument. For example, if it is a multimeter, specify the maximum to minimum values of the voltages, current and resistance to be measured. For instruments that are frequency-sensitive, the frequency range of operation should be specified. For frequency counters,

the minimum to maximum frequency to be measured should be specified. The requirement may be from a few hertz to a few gigahertz depending on the application.

In some cases, both frequency and amplitude ranges have to be specified. Signal generators and frequency synthesisers are such types, and one has to specify the output frequency range and maximum power output levels. Power meter is another such instrument where one has to specify the range of frequencies for which power has to be measured and the range of power level to be measured. The measurement range needed may be from a few microwatts to a few watts. The ranges may be covered with different power sensors.

Accuracy. Typically, the term 'accuracy' in measurements refers to the closeness of a measurement result to the true value. The cost of the instrument increases with increase in its accuracy level.

The requirement of accuracy level depends upon the nature of measurement. For example, laboratories engaged in calibration and issuing certificates need highly accurate instruments. A company that makes consumer articles need not have very accurate instruments.

Let's consider another company that supplies crystal oscillators having stability of the order of $1E-7$ /day. Naturally, the counter used by the company should be at least ten times more stable than this. The measurement accuracy of the frequency counter depends upon the stability of the crystal oscillator used as reference for time-base generation in the counter.

It may use an oven-controlled crystal oscillator or temperature-compensated crystal oscillator. Counters having stability of $2E-8$ /day to $5E-10$ /day are available and you can select a particular one depending upon the purpose of measurement. Basic counters have a low order of stability but offer options for higher-order stability. One has to mention the option in case a higher stability is required. For exam-

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